

Diploma in

PHOTOSHOP



SUMMARY NOTES

Black and White Editing

Lesson 7

CONVERTING IMAGES FROM COLOUR, INTO STRONG DYNAMIC BLACK AND WHITE IMAGES IS NO EASY TASK. YOU WILL LEARN HOW TO MANIPULATE COLOURS IN AN IMAGE, TO CREATE STUNNING TONES AND CONTRAST WHEN THEY ARE CONVERTED INTO BLACK AND WHITE. THIS WILL ENSURE YOU ARE PRODUCING IMAGES THAT RESEMBLE THE WORK OF THE TRADITIONAL MASTERS OF BLACK AND WHITE PHOTOGRAPHY.

5.01	Black & White
5.02	Basic Adjustments
5.03	HSL
5.04	Split Tones
5.05	Vignette

COLOUR TERMINOLOGY

Warm Colours

Warm colours include red, orange, and yellow, and variations of those three colours. These are the colours of fire, of autumn leaves, and of sunsets and sunrises, and are generally energizing, passionate, and positive. Red and yellow are both primary colours, with orange falling in the middle, which means warm colours are all truly warm and aren't created by combining a warm colour with a cool colour. Use warm colours in your designs to reflect passion, happiness, enthusiasm, and energy.

Red is a very hot colour.

It's associated with fire, violence, and warfare. It's also associated with love and passion. In history, it's been associated with both the Devil and Cupid. Red can actually have a physical effect on people, raising blood pressure and respiration rates. It's been shown to enhance human metabolism, too. Red can be associated with anger, but is also associated with importance (think of the red carpet at awards shows and celebrity events). Red also indicates danger (the reason stop lights and signs are red, and that most warning labels are red). Outside the western world, red has different associations. For example, in China, red is the colour of prosperity and happiness. It can also be used to attract good luck. Be aware of cultural differences!

In design, red can be a powerful accent colour. It can have an overwhelming effect if it's used too much in designs, especially in its purest form. It's a great colour to use when power or passion want to be portrayed in the design. Red can be very versatile, though, with brighter versions being more energetic and darker shades being more powerful and elegant.

Alternate meanings: In some Eastern cultures, red symbolizes good fortune and prosperity and is the colour worn by brides on their wedding day. Worldwide, red has been associated with various political movements and has symbolized revolution. In branding: Red often communicates strength, confidence, and power and is a highly visible colour.

Yellow is often the brightest & most energizing of the warm colours.

It's associated with happiness and sunshine. Yellow can also be associated with deceit and cowardice, though (calling someone yellow is calling them a coward). Yellow is also associated with hope, as can be seen in some countries when yellow

ribbons are displayed by families who have loved ones at war. Yellow is also associated with danger, though not as strongly as red. In some countries, yellow has very different connotations. In Egypt, for example, yellow is for mourning. In Japan, it represents courage, and in India it's a colour for merchants.

In your designs, bright yellow can lend a sense of happiness and cheerfulness. Softer yellows are commonly used as a gender-neutral colour for babies (rather than blue or pink) and young children. Light yellows also give a more calm feeling of happiness than bright yellows. Dark yellows and gold-hued yellows can sometimes look antique and be used in designs where a sense of permanence is desired. Alternate meanings: In some Eastern and Asian cultures, yellow is associated with royalty or high rank. In parts of Africa and Latin America, yellow is the traditional colour of mourning.

In branding: Pure/bright yellow does a great job of attracting attention, but can be visually disturbing or even hard to see (for instance, white text against a bright yellow background or vice versa) if not used with care.



COLOUR SCHEMES

Colour context refers to how we perceive colours as they contrast with another colour. The way in which we use two colours together changes how we perceive it. So when you're choosing colours for your graphic designs, think about how much contrast you want throughout the design.

Monochromatic colour schemes are made up of different tones, shades and tints within a specific hue. These are the simplest colour schemes to create, as they're all taken from the same hue, making it harder to create a jarring or ugly scheme (though both are still possible!). Although it lacks colour contrast, it often ends up looking very clean and polished. It also allows you to easily change the darkness and lightness of your colours.



Complementary schemes are created by combining colours from opposite sides of the colour wheel. In their most basic form, these schemes consist of only two colours, but can easily be expanded using tones, tints, and shades. The complementary colour scheme provides the greatest amount of colour contrast. Because of this, you should be careful about how you use the complementary colours in a scheme. It's best to use one colour predominantly and use the second colour as accents in your design.



Analogous schemes are created by using three colours that are next to each other on the 12-spoke colour wheel. Generally, analogous colour schemes all have the same Chroma level, but by using tones, shades and tints we can add interest to these schemes and adapt them to our needs for designing websites.

Analogous structures do not create themes with high contrasting colours, so they're typically used to create a softer, less contrasting design.



Triadic schemes are made up of hues equally spaced around the 12-spoke colour wheel. This is one of the more diverse colour schemes. Triadic colour schemes offer high contrasting colour schemes while retaining the same tone, but they can also seem overpowering if all of your colours are chosen on the same point in a line around the colour wheel. To subdue some of your colours in a triadic scheme, you can choose one dominant color and use the others sparingly, or simply subdue the other two colours by choosing a softer tint.



Split complementary, instead of using colours that are opposites, you use colours on either side of the hue opposite your base hue. A split complementary scheme includes one dominant colour and the two colours directly adjacent to the dominant colour's complement. This created more nuanced colour palette than complementary colour scheme while still retaining the benefits of contrasting colours.



The split complementary colour scheme can be difficult to balance well because unlike analogous or monochromatic colour schemes, the colours used all provide contrast (similar to the complementary scheme). The positive and negative aspect of the split complementary colour model is that you can use any two colours in the scheme and get great contrast but that also means it can also be tricky to find the right balance between the colours. As a result, you may end up playing around with this one a bit more to find the right combination of contrast.

COLOUR SYSTEMS

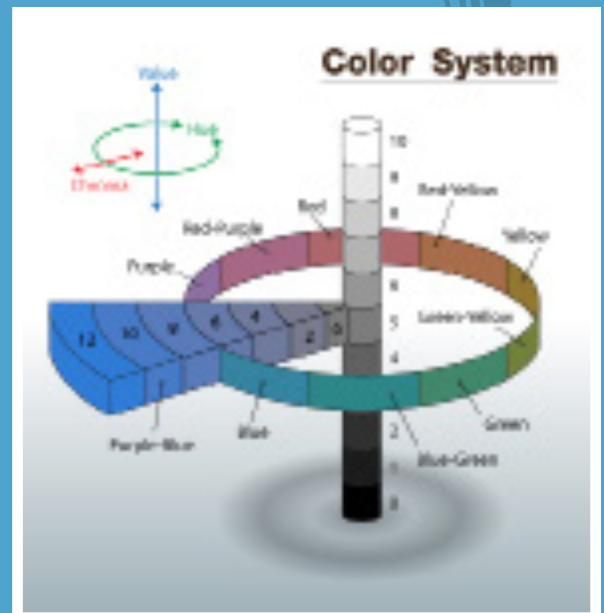
Orange is a very vibrant and energetic colour.

In its muted forms, it can be associated with the earth and with autumn. Because of its association with the changing seasons, orange can represent change and movement in general. Because orange is associated with the fruit of the same name, it can be associated with health and vitality. In designs, orange commands attention without being as overpowering as red. It's often considered more friendly and inviting, and less in-your-face.

Alternate meanings: In India, saffron, a certain yellowish shade of orange, is considered sacred. In Japan, orange is the colour that symbolizes love. In branding: Orange often represents youthfulness and creativity. Gold, which is a type of orange or yellow depending on its hue, is a symbol of luxury or high quality.

Hue is the most basic of colour terms and basically denotes an object's colour. When we say "blue," "green" or "red," we're talking about hue. The hues you use in your designs convey important messages as we have just looked at. Using a lot of pure hues together can add a fun and playful look to a design. Combining high and low saturation in the same hue can make for a sophisticated and elegant design.

Chroma refers to the purity of a colour. A hue with high Chroma has no black, white or grey in it. Adding white, black or grey reduces its Chroma. It's similar to saturation but not quite the same. Chroma can be thought of as the brightness of a colour in comparison to white. In design, avoid using hues that have a very similar Chroma. Opt instead for hues with chromes that are the same or a few steps away from each other. Colours with very high Chroma are best used in moderation, as done here. Differences in Chroma can make for a visually pleasing gradient.



Saturation refers to how a hue appears under particular lighting conditions. Think of saturation in terms of weak vs. strong or pale vs. pure hues. In design, colours with similar saturation levels make for more cohesive-looking designs. As with Chroma, colours with similar but not identical saturations can have a jarring effect on visitors.

Value could also be called “lightness.” It refers to how light or dark a colour is. Lighter colours have higher values. For example, orange has a higher value than navy blue or dark purple. Black has the lowest value of any hue, and white the highest. When applying colour values to your designs, favour colours with different values, especially ones with high Chroma. High contrast values generally result in more aesthetically pleasing designs.

Tones are created when grey is added to a hue. Tones are generally duller or softer-looking than pure hues. Tones are sometimes easier to use in designs. Tones with more grey can lend a certain vintage feel to websites. Depending on the hues, they can also add a sophisticated or elegant look.

Tints, A tint is formed when white is added to a hue, lightening it. Very light tints are sometimes called pastels, but any pure hue with white added to it is a tint. Tints are often used to create feminine or lighter designs. Pastel tints are especially used to make designs more feminine. They also work well in vintage designs and are popular on websites targeted at parents of babies and toddlers.

SUMMARY

Hue is colour (blue, green, red, etc.).

Chroma is the purity of a colour (a high Chroma has no added black, white or grey).

Saturation refers to how strong or weak a colour is (high saturation being strong).

Value refers to how light or dark a colour is (light having a high value).

Tones are created by adding grey to a colour, making it duller than the original.

Shades are created by adding black to a colour, making it darker than the original.

Tints are created by adding white to a colour, making it lighter than the original.

Munsell's Colour System.

There are 3 primary ways to describe a colour. Temperature could be a 4th.
Hue — another word for colour
Saturation (Chroma) — the intensity or purity of a hue
Lightness (value) — the relative degree of black or white mixed with a given hue
Temperature — the perceived warmth or coolness of a colour

Munsell's system forms the basis of the familiar HSL and HSB colour spaces we find in graphics editors. Munsell's system divides colour into three dimensions, which you can see in the image.

- Hue — is measured by degrees around a horizontal circle.
- Chroma — is measured radially from the centre of the circle. The further from the centre the more pure the hue.
- Value / Lightness — runs perpendicular through the centre of the circle. The lowest point is black and the highest point is white with shades of grey at all points between.

You might recognize the system as hue, saturation, and lightness or hue, saturation, and brightness. The values in Munsell's system are different, though. Munsell's hue circle is divided into 10 equal divisions, Chroma is measured from 1–15, and value is on a scale from 1–11. The basic dimensions are in place and the system is internationally accepted and used by the U.S. Bureau of Standards. In Munsell's system there are 5 primary colours, red, yellow, green, blue, and purple. There are also 5 intermediary colours between the primaries, red-yellow, yellow-green, and so on. Colours of equal Chroma and value and at opposite ends of a hue circle are complementary. Adding them creates a neutral grey (the centre of the circle) of the same value.

HSB

HSB (also HSV) is a way to define colour based on how we describe it. This is the one that uses a colour wheel. It came in when we got on-screen colour, as a way to try to describe colour more intuitively than with RGB numbers. HSV stands for hue, saturation, and value, and is also often called HSB (B for brightness). Based on the human perception of colour, the HSB model describes three fundamental



BLACK & WHITE

“There is a timelessness and drama to black and white photography that has meant even with the introduction of colour photography and now digital photography it has remained incredibly popular with photography enthusiasts.”

There is a timelessness and drama to black and white photography that has meant even with the introduction of colour photography and now digital photography it has remained incredibly popular with photography enthusiasts.

However it is important to start thinking in Black and white and to understand that shooting in black and white successfully is a lot more than simply changing a colour photo to a black and white photo. With colour removed we can begin to break down the elements of a photograph. Options like tone, lighting, contrast, texture, form, shape and so on become more important as we cannot hide them behind colour as such. These are the elements that we can use to create interesting and dramatic black and white photos no matter what subject matter you are photographing.

In the first part of this lesson we looked at what makes a good Black and White photo at the time of exposure. This second part of the lesson will look at making a stunning Black and White conversion.

Keep in mind that this lesson is practical based and will need to be reviewed to fully harness the education.

BASIC ADJUSTMENTS

Exposure Options

Exposure adjustments are simple with a RAW file. Use the histogram on the top right to guide your choices, pay close attention to the extreme right and extreme left to ensure that your tones are not clipping.

1. **Exposure** slider will adjust the image's overall exposure.
2. **Contrast** will boost overall image contrast in both the shadows and highlights at the same time.
3. **Highlights** slider will help recover some of the details lost in highlights.
4. **Shadows** slider will help lift some of the shadows.
5. **Whites** looks specifically at the overall bright tones in the image.
6. **Blacks** looks at the overall dark tones in the image.
7. **Clarity** adds mid-tone contrast to your image, making it look sharper, even though it doesn't actually add sharpening. Adds visual dynamic to image.
8. **Vibrance** changes the saturation of all lower-saturated colours with less effect on the higher saturated colours. Vibrance also prevents skin tones from becoming oversaturated.
9. **Saturation** adjusts the saturation of all image colours equally.

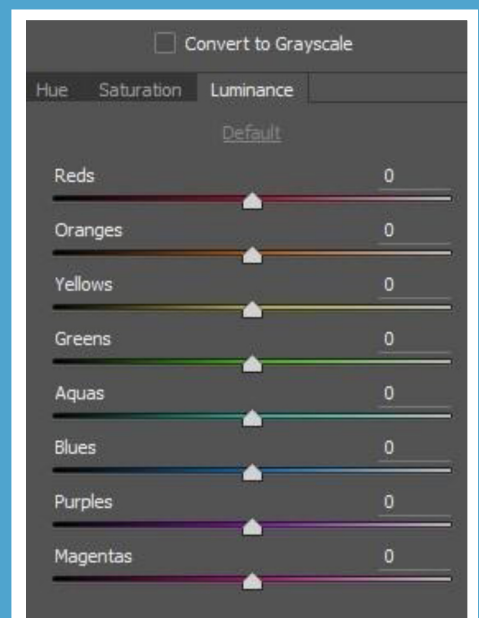


HSL stands for “Hue,” “Saturation,” and “Luminance,” which you can see when you expand the panel. Hue deals with the actual colour tones of the image, like shifting colour tones to a different tonal range. Saturation deals with the power or the saturation of a specific colour while Luminance deals with the brightness of a certain colour in the image.

Hue is the actual colour. We have different Hues in our images and when we adjust the Hues sliders we will affect the tonal range of each colour. The Hue sliders let you replace colours in your photo with neighbouring hues from the colour wheel. For example, if we want to affect the Blues in our image, we can drag the slider for Blue to the left or right. Dragging the slider to the left will make the Blues in our image greener, whereas dragging the slider to the right will increase the purple in our image.

Saturation is the intensity or purity of a hue. Similarly to the Hue sliders, we can use the slider to change the look of our image. When we have the Saturation sliders selected, any colour of the image that we have selected will either increase or decrease in saturation depending on how we adjust the slider.

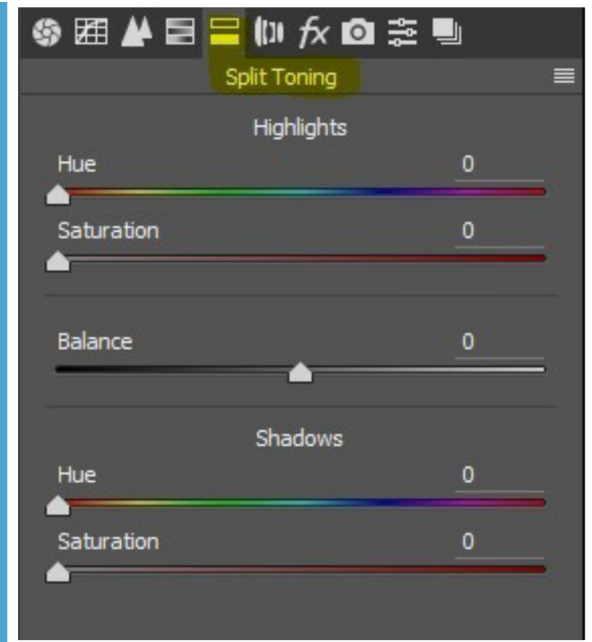
Luminance measures the relative degree of black or white that’s been mixed with a given hue. Adding white makes the colour lighter and adding black makes it darker. The effect of lightness or value is relative to other values in the photo. You can make a colour seem lighter by placing it next to a darker colour. The Luminance deals with the actual brightness levels of the different colours. You can use this tool to increase the tonal separation of the colours and add contrast as well. Down below you can see the before- after result.



SPLIT TONES

Split Toning will allow you to add to colours to a photo to tone it. These can be 2 of the same colours or 2 different colours and the can be applied to a colour or a Black and White Photo. We will look at this in relation to Black and White toning.

In order to add some colour to the Highlights and Shadows, make sure to use the adjustment sliders and click on the Highlights button to select your Hue and the level of saturation. Alternatively, you can press ALT + CLICK (PC) or COMMAND + CLICK (MAC) on the Hue slider to have a preview of the colour that you are going to apply.



VIGNETTE

The vignette will help you to give a different look to your image and create a centre of attention on the subject, depending on the settings. You can always start with the Lens Correction panel, click Manual and use the Lens Vignetting slider to increase or decrease the amount of vignetting. If however, you have significantly cropped the image, then you may need to select the Effects panel and apply the Post-Crop Vignetting, because the Post-Crop Vignetting Effects vignette is applied to the image after cropping.

There are three Styles:

1. **HighlightPriority** enables highlight recovery but can lead to colour shifts in darkened areas of a photo. It is suitable for photos with bright image areas such as clipped specular highlights and behaves more like a traditional exposure burn.
2. **ColourPriority** minimizes colour shifts in darkened areas of a photo but cannot perform highlight recovery. Also behaves more like a traditional exposure burn.
3. **PaintOverlay** is similar to an overlay of black or white paint.

